Supplementary Material for From Mechanism Activation to Forecast Value

1 Purpose and scope

This supplement preserves the study-level detail behind the synthesis. It does not pool task-specific scores. Each figure is copied from the corresponding returned evidence archive. The figures are descriptive mechanism and factorial diagnostics; the numeric tables and dated inference in the main paper determine the claims.

2 Evidence inventory

Study	Run identifier	Fits	Preserved evidence
C3R	RUN-20260929-082738-C3R	33	Source and data hashes; checkpoints; dated RankIC; paired and factorial inference; coalition and intervention diagnostics.
DARC	RUN-20260929-120135-DARC	18	Source and data hashes; checkpoints; frozen forecast streams; eight readouts; paired, factorial, and economic diagnostics.
CARE	RUN-20260930-064205-CARE	24	Source and data hashes; quantile predictions; calibration metrics; paired and factorial inference; analog and routing diagnostics.
MOSAIC	RUN-20261002-123004-MOSAIC	21	Source and data hashes; variance forecasts; paired and factorial inference; branch, stress, and exposure diagnostics.
DASH-V	RUN-20261002-151054-DASHV	15	Source and data hashes; frozen forecasts; delayed replay; paired and factorial inference; pooling and memory diagnostics.

3 Mechanism-to-claim register

The five systems expose different internal objects, so a common diagnostic cannot be imposed by name alone. Table 2 records the observable used to establish activation, the matched counterfactual used to test incremental value, and the principal empirical bottleneck. This mapping prevents a mechanism plot from being interpreted as a performance comparison.

Table 2: Claim map for the five mechanisms. "Activation" identifies an internal change; "matched test" identifies the forecast-level counterfactual.

Study	Activation evidence	Matched forecast test	Main diagnostic bottleneck
C3R	Message energy, coalition inclusion, and routing entropy	Full communication-and-risk router versus the plain and risk-only cells	Candidate risks were compressed and routing remained diffuse
DARC	Forecast disagreement, pairwise corrections, and gate adoption	Pair-specific arbitration and gating versus global or fixed readouts	Local pair decisions did not reliably improve the completed ranking
CARE	Different analog sets, memory disagreement, and route entropy	Dual-memory tail route versus single-memory and dual-plain routes	Retrieval changed representations without consistent pinball-loss gain
MOSAIC	Nonzero downside and market-stress branch outputs	Two-branch full model versus anchor-only and one-branch cells	Corrections improved a weak anchor family but not strong controls
DASH-V	Time-varying pool forecasts, local reliability, and memory weights	Hierarchical multi-memory pool versus static, local, and single-memory pools	Shrinkage and near-equal memory allocation limited adaptive differentiation

4 Decision-theoretic mechanism audit

The common structure can be expressed as a three-stage value chain. Let b_t be the matched forecast at date t, let d_{tk} denote the change associated with candidate action k, and let q_{tk} be the weight assigned using the information available at the forecast origin. The deployed change is $d_t = \sum_k q_{tk} d_{tk}$. For a smooth convex loss, the local improvement over the matched forecast is organized by

$$\Delta \approx -\,\mathrm{E}[g_t d_t] - \tfrac{1}{2}\,\mathrm{E}[h_t d_t^2], \qquad g_t = \partial_b \ell(y_t, b_t), \quad h_t \geq 0.$$

The first term measures alignment with avoidable future error. The second is the curvature cost of moving the forecast. Selection error, delayed outcome feedback, and transfer from validation to test further reduce the deployable gain. The expression is therefore a diagnostic decomposition: activity raises $|d_t|$, but forecast value requires the signed alignment term to dominate the curvature and deployment costs.

This yields three empirical restrictions. First, the candidate actions must be *separable*: experts, memories, branches, or pooling rules must produce meaningfully different decisions. Second, the selector must be *loss-aligned*: observed disagreement must predict which action will avoid future loss. Third, the relation must be *deployable*: it must survive finite validation information, label maturity, and the later test regime. The main article's matched reductions and factorial cells test these transitions; the figures below show the internal quantities that explain where a transition weakened.

Table 3: Model-specific location of the limiting transition. The mapping links an observed diagnostic to the theory of forecast value and to a targeted redesign.

Study	Intended source of value	Limiting transition in the returned evidence	Design implication
C3R	Communication patterns select lower-risk expert coalitions	Candidate coalition losses were too compressed to create a separated routing problem	Rescale the routing target or first establish material coalition-loss dispersion; adding more messages cannot repair an uninformative target
DARC	Local pairwise advantage improves the completed asset ranking	Pairwise signals operated, but their gains dissipated through nonlinear rank completion	Train and validate the arbitration rule against the final date-level ranking; include an adoption-matched fixed gate
CARE	Two memories identify which conditional-quantile component needs adjustment	Memories differed, but their disagreement was not aligned with the later component-specific pinball gradient	Define a prospective condition under which one memory has lower component loss before adding a more flexible route
MOSAIC	Stress branches correct a persistent variance anchor in adverse states	Branches improved the anchor, while the forecast family remained below strong external controls	Strengthen or replace the anchor first; then test whether branch information adds value beyond an equally informed strong base
DASH-V	Matured losses reveal persistent changes in relative model quality	Delay, shrinkage, and weak loss separation contracted the hierarchy toward near-equal memory allocation	Verify persistent rank changes in the frozen forecast bank and calibrate adaptation against a static pool at matched information

4.1 Why the five failure regimes are economically distinct

C3R: action separability. Coalition routing can create value only if alternative communication patterns lead to materially different expected losses. When the risk targets are nearly indistinguishable relative to the normalization scale, a regularized router is pulled toward diffuse weights. The resulting forecast can be numerically stable, yet the selector has little economically relevant choice. This is a representation bottleneck: the downstream route cannot recover information that the candidate actions do not express.

DARC: aggregation and non-translation. Pairwise correctness is not additive. Local preferences can form cycles, cancel across assets, or change the completed ranking in positions that carry little date-level score. A useful arbitration layer must therefore improve the final ordering, not merely classify many pairs correctly. Comparing the learned gate with a fixed gate at the same adoption rate separates information in the gate from the mechanical effect of allowing more interventions.

CARE: disagreement without loss alignment. Two analog memories can retrieve different histories and still be equally uninformative about which conditional-quantile component is wrong. The decision-relevant signal is the covariance between the memory difference and the future component-specific pinball gradient. Ordered output quantiles remove one mechanical failure mode, so the remaining comparison focuses on whether retrieval diversity identifies avoidable distributional error.

MOSAIC: local correction versus family adequacy. An additive branch can improve its own anchor while leaving a large residual gap to an external forecast family. This is a bias hierarchy: the correction addresses a particular asymmetry, whereas the anchor may still omit more basic dynamics. Within-family improvement is meaningful evidence about the branch, but adoption requires the corrected family to clear the maintained external benchmark.

DASH-V: delayed adaptation and contraction. Adaptive pooling relies on persistent differences in matured forecast loss. Short memories contain little effective evidence after a multi-date target matures; long memories are

more stable but average across potential regime changes. Hierarchical shrinkage reduces estimation variance by moving local weights toward shared values. When relative losses are weak or unstable, the rational consequence is contraction toward static pooling, which limits both risk and attainable gain.

5 Reviewer-oriented evidence map

The evidence package assigns one question to each artifact rather than asking the reader to infer performance from visual complexity. This ordering also makes the audit reproducible: a reviewer can begin with the forecast-level claim and then inspect the mechanism diagnostics that explain it.

Table 4: Evidence hierarchy used in the main paper and supplement.

Evidence layer	Question answered	Interpretation rule
Completion and finite-output records	Did the implemented cell execute and return usable forecasts?	Establishes operational integrity only
Activation diagnostics	Did messages, gates, memories, branches, or pool weights vary as intended?	Establishes that the proposed channel operated
Matched reduction	Did the channel improve forecasts when information and training opportunity were held fixed?	Identifies incremental value within the model family
Four-cell factorial	Did two proposed components contribute separately or complement one another?	Attributes the full-cell contrast and detects redundancy or interference
Strong external controls	Is the resulting family competitive with maintained alternatives?	Determines whether a within-family gain supports practical adoption
Dated block inference and multiplicity control	Is the comparison distinguishable from dependent sampling variation across market dates?	Determines the strength and scope of the empirical claim
Cross-panel retention	Does the interpretation survive all observed markets, including adverse panels?	Prevents a favorable panel from replacing the complete record

Table 5: Returned-evidence completeness. Every item is retained for all three panels.

Study	Fixed config	Source/data hashes	Dated outputs	Paired/factorial inference	Vector figures
C3R	Yes	Yes	Yes	Yes	24
DARC	Yes	Yes	Yes	Yes	24
CARE	Yes	Yes	Yes	Yes	24
MOSAIC	Yes	Yes	Yes	Yes	24
DASH-V	Yes	Yes	Yes	Yes	24

All five status files report completion of every recorded fit. No failed fit is excluded from the synthesis. Mixed and adverse comparisons remain in the study-level CSV and JSON evidence. The fitted-cell total is 111; statistical estimation and frozen readouts are additional.

6 Open-source implementation map

The public release preserves each historical training package as an independent, configuration-fixed program. Table 6 maps the model name used in the manuscript to the visible entry point, the file implementing its principal mechanism, and the compact evidence returned by the corresponding run. This map allows reviewers to distinguish

executable model logic from manuscript-level interpretation. Dataset arrays and raw checkpoints are excluded because of source rights and file size; acquisition records, processed-file hashes, configurations, and adverse result summaries remain in the release.

Table 6: Implementation-to-evidence traceability in the public repository. Paths are relative to the repository root.

Study	Entry point	Principal mechanism	Released evidence
C3R	`model_code/c3r/main.py`	`src/c3rmoe/c3r.py`	Configuration, training and routing code; panel metrics, paired RankIC inference, factorial interaction, and status files
DARC	`model_code/darc/main.py`	`src/darc/council.py`	Frozen-forecast council, pairwise arbitration and gate code; panel metrics, paired RankIC inference, factorial evidence, and status files
CARE	`model_code/care/main.py`	`src/care/models.py`	Dual-memory quantile models and routing; panel metrics, paired pinball inference, split records, factorial evidence, and status files
MOSAIC	`model_code/mosaic/main.py`	`src/mosaic/models.py`	Positive anchor and correction branches; panel metrics, paired QLIKE inference, model budgets, factorial evidence, and status files
DASH-V	`model_code/dashv/main.py`	`src/dashv/aggregate.py`	Delayed partially pooled aggregation; panel metrics, registered and paired QLIKE inference, factorial evidence, and status files

All five packages keep their original `python main.py` entry point and declare dependencies without installing them. The repository-level static validator parses Python and JSON, verifies the release manifest, checks the required model files, and rejects committed arrays, checkpoints, logs, or nested archives. These checks establish release integrity only; the forecast claims in the manuscript remain tied to the returned AutoDL evidence.

7 Panels and chronological splits

Table 7: Raw public panel inventory. Eligible assets vary by date after past-history checks.

Panel	Dates	Symbols	Span	Public source
SP500	3,768	502	2010-01-04–2024-12-20	`https://www.kaggle.com/datasets/andrewmvd/sp-500-stocks`; current-constituent limitation applies
KDD17	2,518	50	2007-01-03–2016-12-30	`https://github.com/fulifeng/Adv-ALSTM`
ACL18	1,258	88	2012-09-04–2017-09-01	`https://github.com/yumoxu/stocknet-dataset`

Table 8: Outer chronological splits for C3R and DARC. Counts are forecast origins.

Panel	Segment	First origin	Last origin	Count
SP500	train	2010-03-02	2018-12-20	2,219
	validation	2019-01-02	2020-12-22	499
	test	2021-01-04	2024-12-12	993
KDD17	train	2007-03-01	2014-12-22	1,969
	validation	2015-01-02	2015-12-22	246
	test	2016-01-04	2016-12-21	246
ACL18	train	2012-10-31	2015-12-22	792
	validation	2016-01-04	2016-06-22	119
	test	2016-07-01	2017-08-17	285

Table 9: CARE chronological fit, checkpoint, validation, and test segments.

Panel	Segment	First origin	Last origin	Count
SP500	fit	2010-03-02	2017-08-16	1,880
	checkpoint	2017-08-25	2018-12-20	333
	validation	2019-01-02	2020-12-22	499
	test	2021-01-04	2024-12-12	993
KDD17	fit	2007-03-01	2013-10-10	1,667
	checkpoint	2013-10-21	2014-12-22	296
	validation	2015-01-02	2015-12-22	246
	test	2016-01-04	2016-12-21	246
ACL18	fit	2012-10-31	2015-06-25	667
	checkpoint	2015-07-07	2015-12-22	119
	validation	2016-01-04	2016-06-22	119
	test	2016-07-01	2017-08-17	285

Table 10: MOSAIC and DASH-V chronological segments.

Panel	Segment	First origin	Last origin	Count
SP500	fit	2010-03-31	2017-08-23	1,864
	checkpoint	2017-08-31	2018-12-21	330
	validation	2019-01-02	2020-12-23	500
	test	2021-01-04	2024-12-13	994
KDD17	fit	2007-03-30	2013-10-17	1,651
	checkpoint	2013-10-25	2014-12-23	293
	validation	2015-01-02	2015-12-23	247
	test	2016-01-04	2016-12-22	247
ACL18	fit	2012-11-30	2015-07-02	651
	checkpoint	2015-07-13	2015-12-23	116
	validation	2016-01-04	2016-06-23	120
	test	2016-07-01	2017-08-18	286

8 Comparator and inference register

Study	Recorded comparator family	Inference and multiplicity
C3R	GRU, StockMixer, MASTER, MIGA, MERA, MAFS; graph-only, credit-only, plain mixture, and unprojected C3R cells.	5,000 chronological moving-block draws; test block 20; ten full-model comparisons per panel adjusted by Holm; factorial separate.
DARC	The same six adapted neural controls; equal rank, static weighted, recent performance, uniform pair vote, global always, pairwise always, and global gated readouts.	5,000 chronological moving-block draws; test block 20; validation block 20 except ACL18 block 14; thirteen full comparisons per panel and split adjusted by Holm; factorial separate.

Study	Recorded comparator family	Inference and multiplicity
CARE	Historical and market analogs; GRU, StockMixer, MASTER, and MERA quantile adaptations; single-plain, dual-plain, and single-tail cells.	5,000 chronological moving-block draws; test block 20; validation block 20 except ACL18 block 14; nine comparisons per panel and split, plus 27 cross-panel comparisons per split, adjusted by Holm.
MOSAIC	Persistence, EWMA, HAR, SHAR, market-SHAR, market-SHAR interaction, flat MLP, GRU, RG-ResMoE adaptation; base, own, and market cells.	5,000 chronological moving-block draws; test block 20; validation block 20 except ACL18 block 15; twelve comparisons per panel and split, plus 36 cross-panel comparisons per split, adjusted by Holm.
DASH-V	Persistence, EWMA, HAR, SHAR, market-SHAR, flat MLP, GRU, LSTM, TCN, Patch Transformer; equal geometric, equal arithmetic, static QLIKE, asset-local single-memory; base, hierarchy, and multi-memory cells.	5,000 chronological moving-block draws; test block 20; seventeen panel comparisons; five recorded main hypotheses and fifteen cross-panel hypotheses also adjusted by Holm.

For DARC, the original central report also compared the full readout with equal-rank pooling: 0.000106 [−0.000096, 0.000306] on SP500, −0.002392 [−0.007470, 0.000099] on KDD17, and −0.000254 [−0.001163, 0.000236] on ACL18. For DASH-V, the validation-selected raw forecasts were GRU, Patch Transformer, and GRU; full-model effects were −0.004751, 0.000484, and −0.011058. Effects against static QLIKE pooling were 0.004466, −0.000051, and −0.000033. These recorded primary comparisons and the retrospective test-best external comparisons answer different questions and are kept separate.

The machine-readable `evidence_summary.csv` and `mechanism_figure_manifest.csv` ship beside the source. The latter lists all 120 returned vector figures with hashes. The fifteen factorial panels below were selected by one rule fixed for this supplement: one four-cell factorial figure for every study–dataset pair. Ten additional SP500 panels—two per study—show the distinct activation diagnostics summarized in Table 2. Raw console logs and the remaining mechanism plots stay in the evidence archives; they are not used as performance evidence and are not added to the article PDF.

9 Activation atlas

This atlas asks whether each proposed mechanism created its intended internal variation. All panels are descriptive and use information available under the recorded forecasting clock. They do not replace common-date loss comparisons.

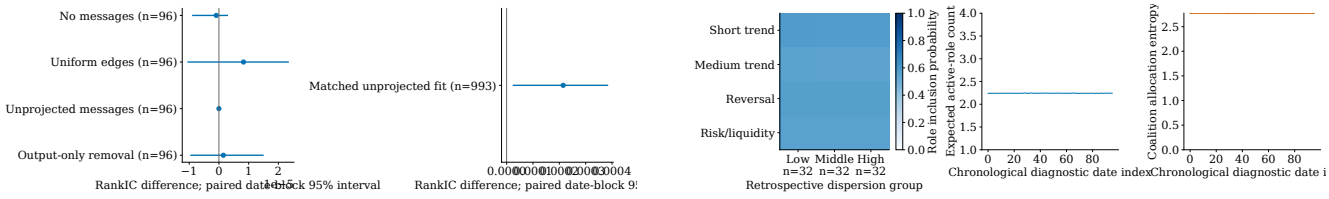


Figure 1: C3R activation on SP500. Left: message interventions and their paired RankIC changes. Right: coalition inclusion, expected active roles, and allocation entropy. The high-entropy, weakly differentiated route is consistent with compressed candidate-risk targets.

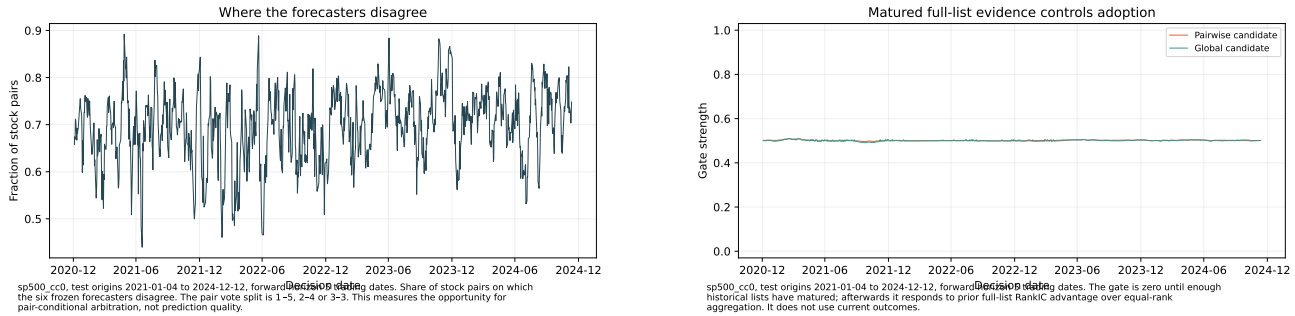


Figure 2: DARC activation on SP500. Forecast disagreement creates opportunities for arbitration (left), while pairwise and global gate strength remain close to one half after the full history matures (right). Gate operation alone does not establish better final rankings.

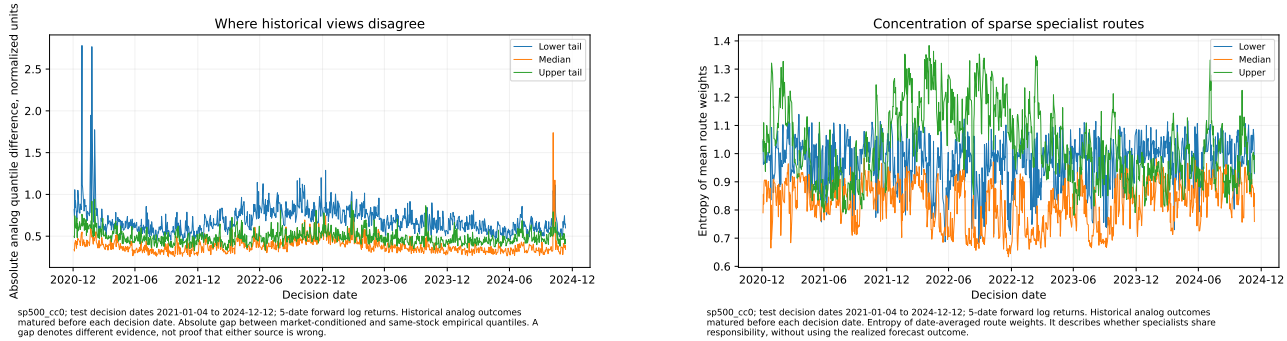


Figure 3: CARE activation on SP500. Stock-specific and market memories disagree by a time-varying amount across quantiles (left), and specialist-route concentration also changes through time (right). The plots establish differentiated retrieval and routing, not lower pinball loss.

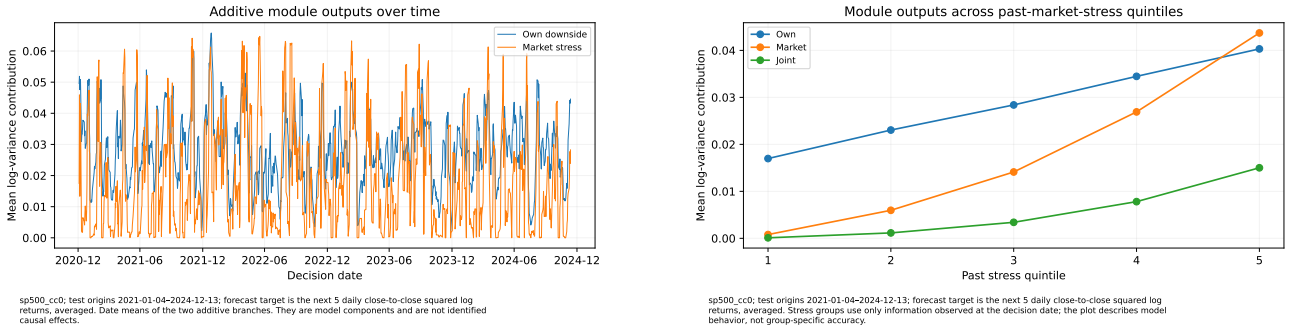


Figure 4: MOSAIC activation on SP500. Both additive correction branches contribute over time (left), and the branch outputs increase across past-stress quintiles in the intended direction (right). These model-behavior diagnostics remain separate from external QLIKE comparisons.

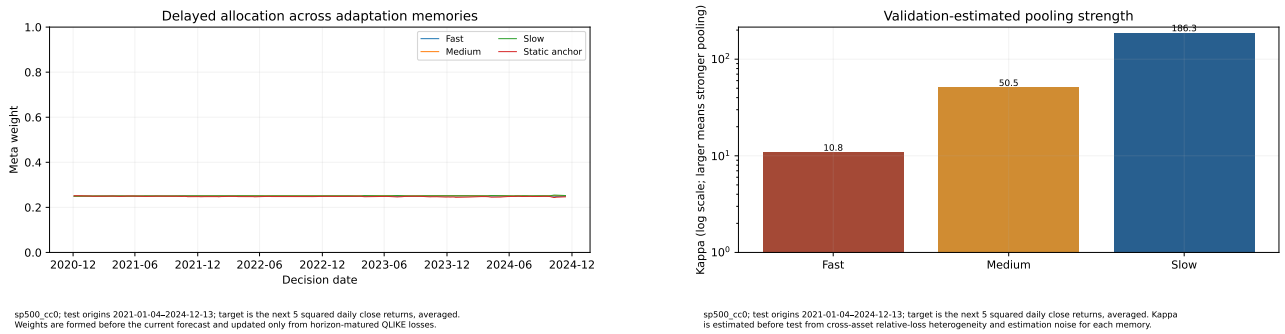


Figure 5: DASH-V activation on SP500. Delayed allocation across fast, medium, slow, and static memories stays close to equal weights (left), while validation-estimated pooling grows with memory scale (right). The hierarchy operates, but adaptive differentiation is limited.

10 Factorial atlas

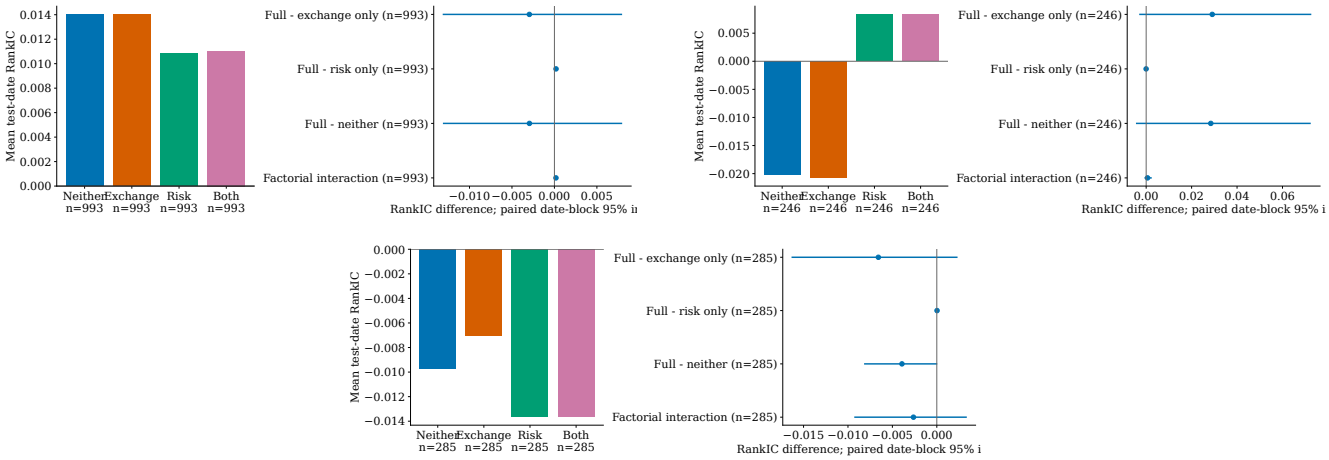


Figure 6: C3R factorial cells on SP500, KDD17, and ACL18. The full model's favorable KDD17 point estimate is not accompanied by interval evidence and is largely reproduced by the risk-only cell.

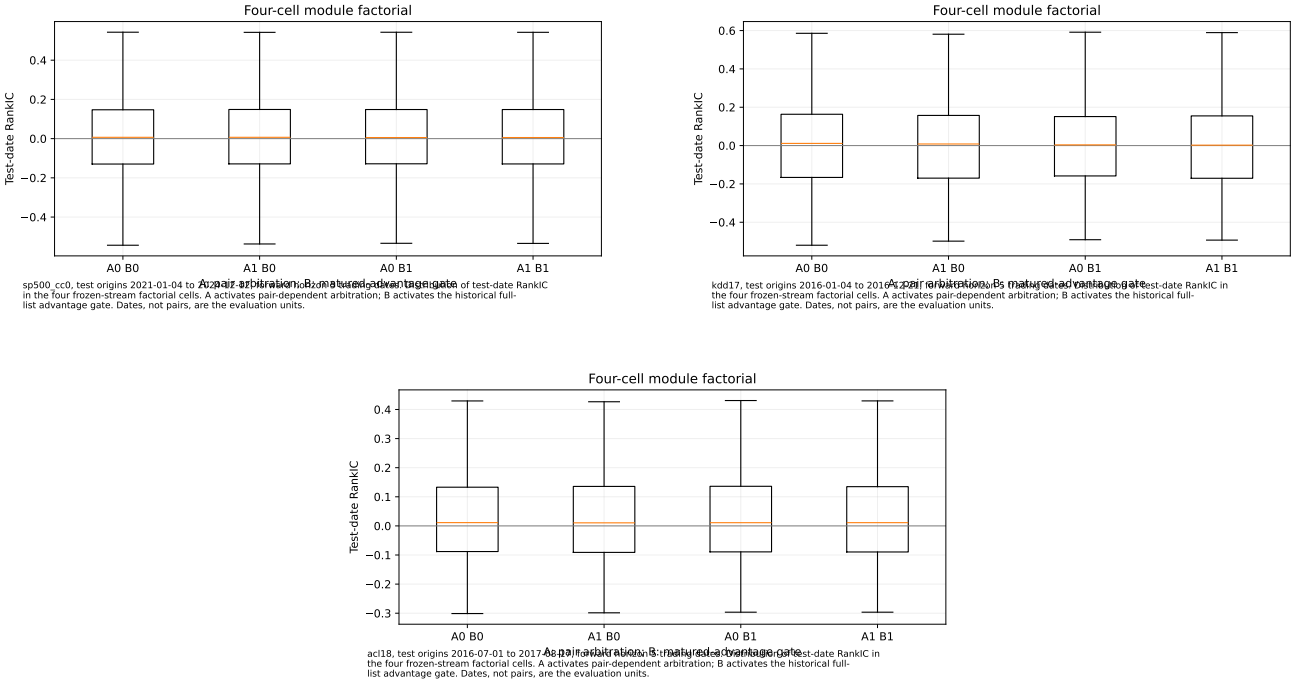


Figure 7: DARC factorial cells on the three panels. Pair-specific arbitration and the matured-advantage gate do not produce a supported interaction.

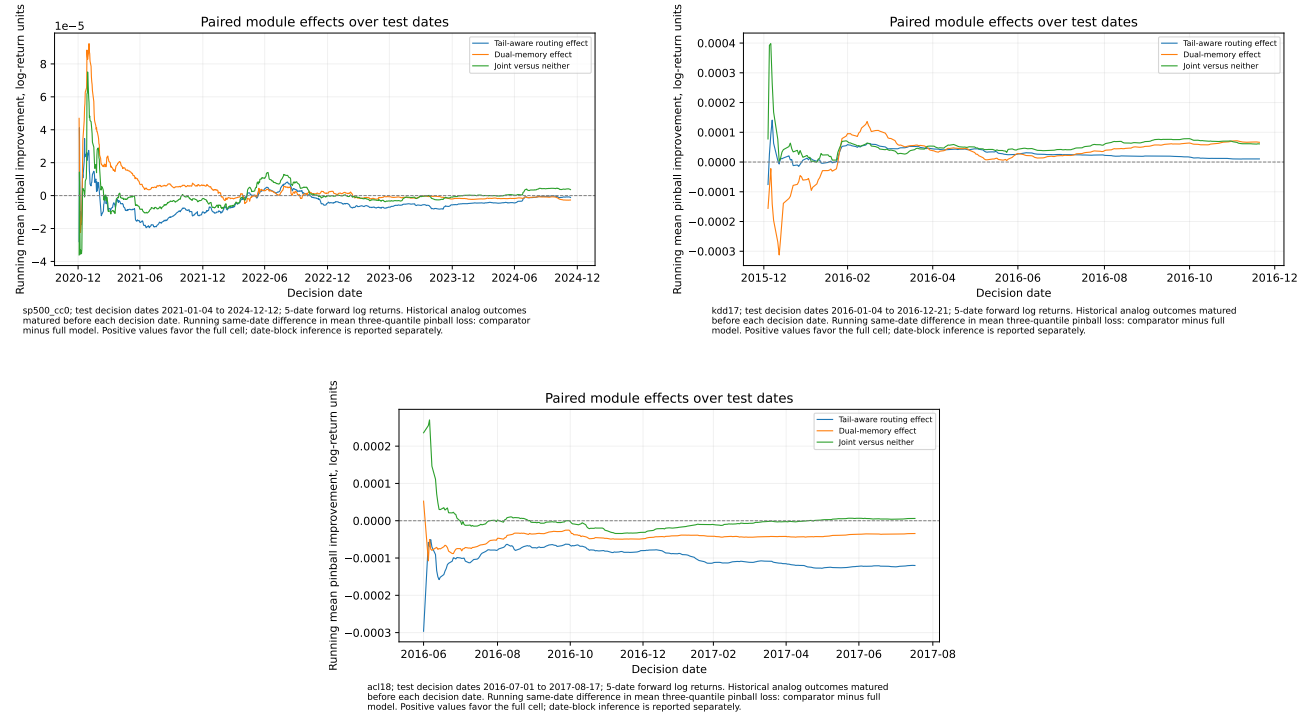


Figure 8: CARE dual-memory by tail-routing factorial. The full cell is competitive but not the best pinball-loss model on any panel.

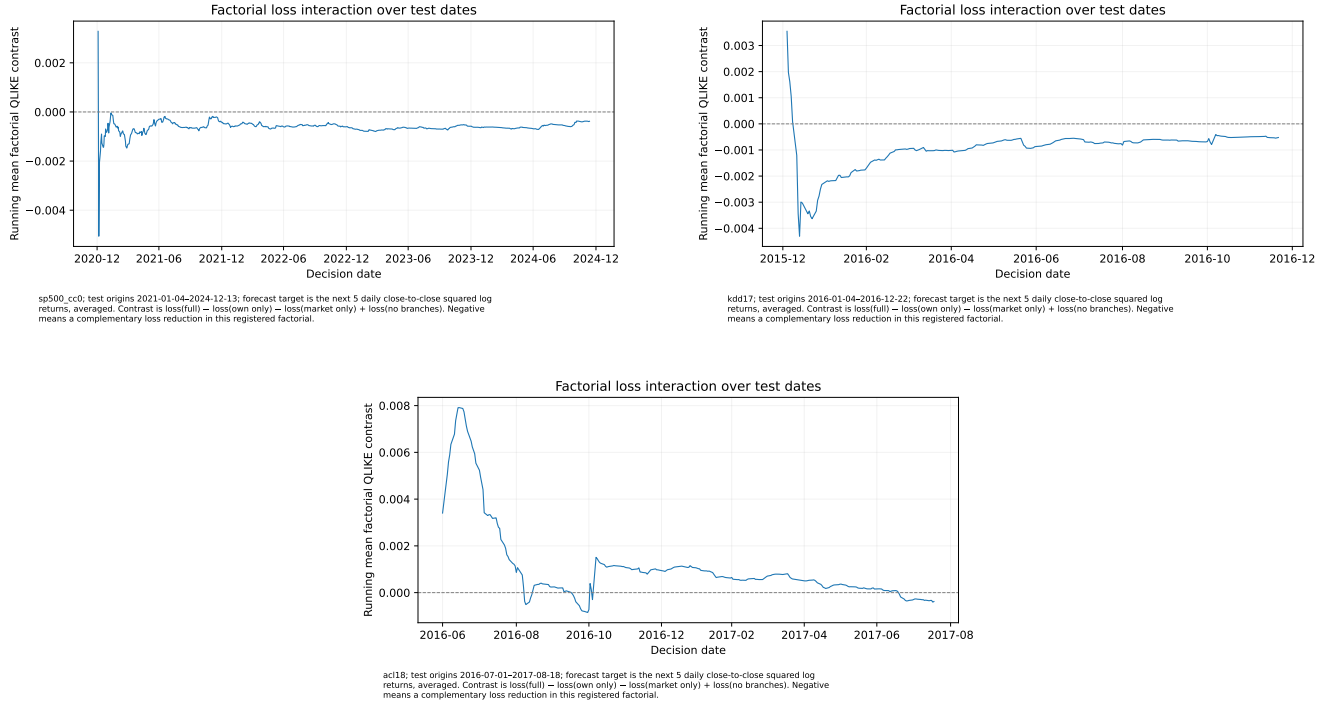


Figure 9: MOSAIC correction-branch factorial. Full-versus-base improvement is clearest on SP500, while the interaction intervals do not establish complementarity.

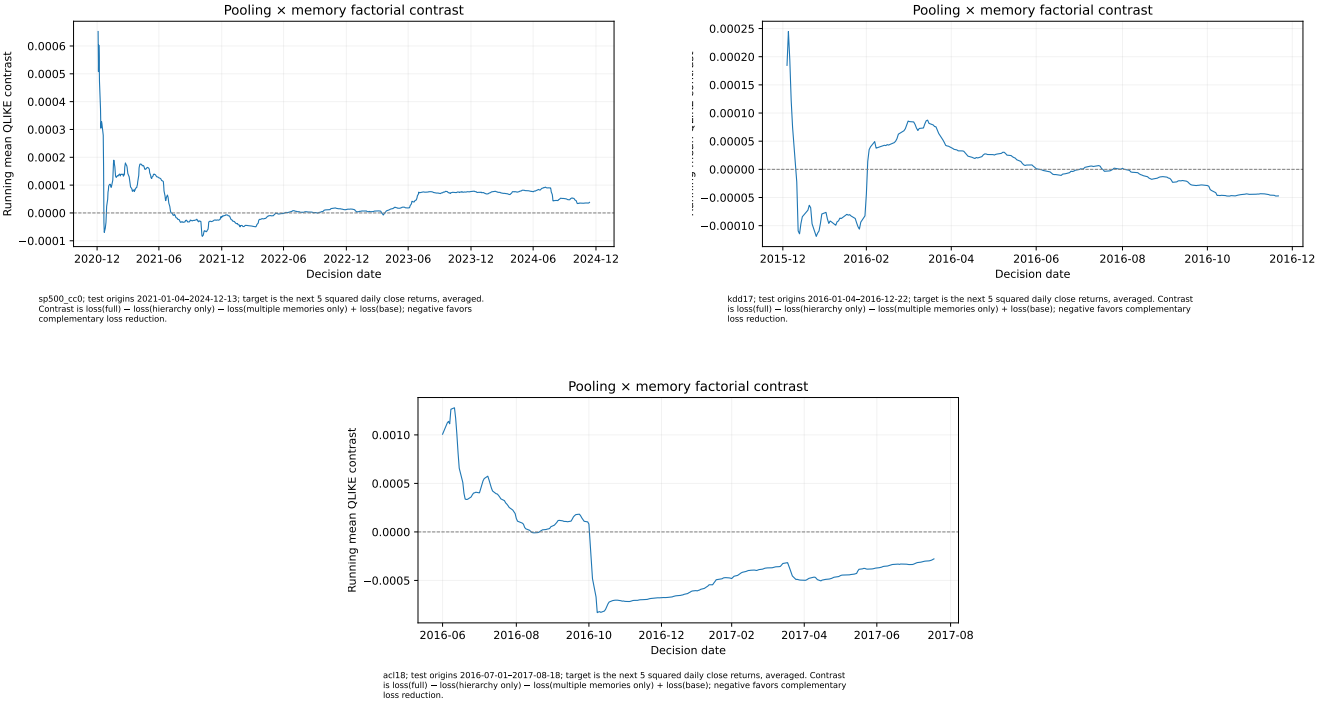


Figure 10: DASH-V hierarchy by multiple-memory factorial. None of the interaction intervals supports the proposed complementarity.

11 Interpretation rules

The atlas is included to make module behavior inspectable. A non-flat route, changing weight, or nonzero correction establishes activation only. Incremental value requires a common-date comparison with the matched reduction. Broad value additionally requires a strong external comparator and consistent evidence across panels. Empty, failed, mixed, and adverse outcomes remain in the underlying archives.

The complete archive contains 120 vector PDF mechanism figures, eight per dataset for each study. Including every plot in the article would obscure the primary comparisons. The supplement therefore shows 25 panels: ten representative activation diagnostics on SP500 and fifteen factorial panels covering every study–dataset pair. The accompanying figure manifest maps every remaining file to its study and dataset, preserves its byte size and hash, and makes the selection auditable.